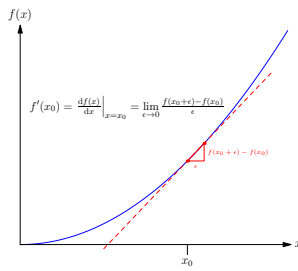


Differential Calculus



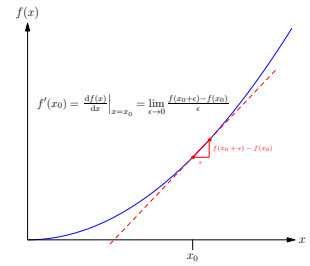
Differentiation, product and chain rules, vectors and matrices

Why Calculus?

- Calculus is a fundamental tool of mathematical analysis
- In machine learning differentiation is fundamental tool in optimisation
- Integration is an essential tool in taking expectations over continuous distributions
- Both differentiation and integration crop up elsewhere

Outline

1. Why Calculus?
2. Differentiation
3. Vector Calculus
4. Matrix Calculus

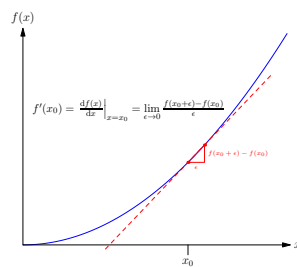


Back to Basics

- You have all done A-level maths so should be familiar with the rules of calculus
- But, it is easy to forget the rules and sometimes we use quite sophisticated tricks
- Although the sophisticated tricks really speed up calculations, it pays to be able to understand where these tricks come from

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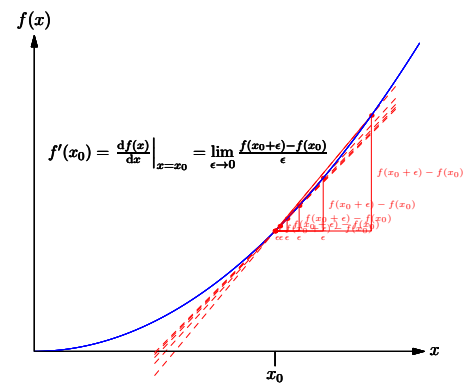
Polynomials

- $f(x) = x^2$

$$\frac{dx^2}{dx} = \lim_{\epsilon \rightarrow 0} \frac{(x+\epsilon)^2 - x^2}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{(x^2 + 2\epsilon x + \epsilon^2) - x^2}{\epsilon} = \lim_{\epsilon \rightarrow 0} 2x + \epsilon = 2x$$
- $(x+\epsilon)^n = (x+\epsilon)(x+\epsilon)\cdots(x+\epsilon) = x^n + n\epsilon x^{n-1} + O(\epsilon^2)$

$$\frac{dx^n}{dx} = \lim_{\epsilon \rightarrow 0} \frac{(x+\epsilon)^n - x^n}{\epsilon} = \lim_{\epsilon \rightarrow 0} n x^{n-1} + O(\epsilon) = n x^{n-1}$$

Differentiation

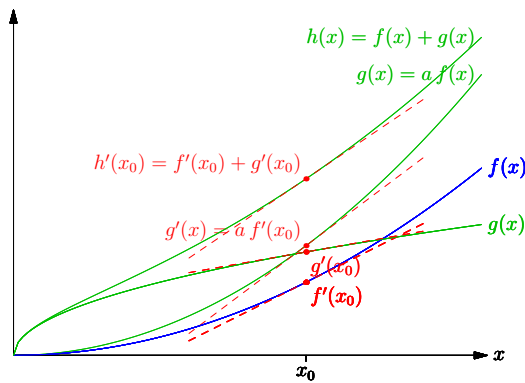


Linearity of derivatives

- Consider the derivative of $af(x) + bg(x)$

$$\begin{aligned} \frac{d(af(x) + bg(x))}{dx} &= \lim_{\epsilon \rightarrow 0} \frac{(af(x+\epsilon) + bg(x+\epsilon)) - (af(x) + bg(x))}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{a(f(x+\epsilon) - f(x)) + b(g(x+\epsilon) - g(x))}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} a \left(\frac{f(x+\epsilon) - f(x)}{\epsilon} \right) + b \left(\frac{g(x+\epsilon) - g(x)}{\epsilon} \right) \\ &= af'(x) + bg'(x) \end{aligned}$$

- Differentiation is a linear operation



Chain Rule

- Recall $f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$
- Let $h(x) = f(g(x))$
- Then

$$\begin{aligned} h(x + \epsilon) &= f(g(x + \epsilon)) = f(g(x) + \epsilon g'(x) + O(\epsilon^2)) \\ &= f(g(x)) + \epsilon g'(x) f'(g(x)) + O(\epsilon^2) \end{aligned}$$

- Thus

$$h'(x) = \lim_{\epsilon \rightarrow 0} \frac{h(x + \epsilon) - h(x)}{\epsilon} = g'(x) f'(g(x))$$

- This is the famous **chain rule**. Together with the product rule it means you can differentiate almost everything

Inverse functions

- Suppose $g(y) = f^{-1}(y)$ is the inverse of $f(x)$ in the sense that $g(f(x)) = f^{-1}(f(x)) = x$
- Using the chain rule

$$\frac{dg(f(x))}{dx} = f'(x) g'(f(x)) = 1$$

since $g(f(x)) = x$

- So $g'(f(x)) = 1/f'(x)$
- Writing $y = f(x)$ so that $x = f^{-1}(y) = g(y)$ we find $g'(y) = 1/f'(g(y))$ that is

$$\frac{dg(y)}{dy} = \frac{1}{f'(g(y))} \quad \frac{df^{-1}(y)}{dy} = \frac{1}{f'(f^{-1}(y))}$$

Functions of Exponentials

- What about $f(x) = e^{cx}$

$$\frac{de^{cx}}{dx} = \frac{de^{cx}}{dcx} \frac{dcx}{dx} = ce^{cx}$$

- More generally using the chain rule

$$\frac{de^{g(x)}}{dx} = g'(x) e^{g(x)}$$

- Also $a^{bc} = (a^b)^c$ (that is we multiply a together $b \times c$ times)

$$\frac{da^x}{dx} = \frac{d(e^{\ln(a)x})}{dx} = \frac{de^{\ln(a)x}}{dx} = \ln(a) e^{\ln(a)x} = \ln(a) a^x$$

- Recall $f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$
- If $h(x) = f(x)g(x)$

$$\begin{aligned} h'(x) &= \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon)g(x + \epsilon) - f(x)g(x)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{(f(x) + \epsilon f'(x) + O(\epsilon^2))(g(x) + \epsilon g'(x) + O(\epsilon^2)) - f(x)g(x)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{\epsilon(f'(x)g(x) + f(x)g'(x)) + O(\epsilon^2)}{\epsilon} = f'(x)g(x) + f(x)g'(x) \end{aligned}$$

- This is the **product rule**

More on chain rules

- We can also write the chain rule as

$$\frac{df(g(x))}{dx} = \frac{df(g)}{dg} \frac{dg(x)}{dx}$$

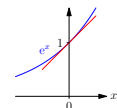
- Sometimes this is neater or easier to remember

$$\begin{aligned} \frac{de^{\cos(x^2)}}{dx} &= \frac{de^{\cos(x^2)}}{d\cos(x^2)} \frac{d\cos(x^2)}{dx^2} \frac{dx^2}{dx} \\ &= e^{\cos(x^2)} (-\sin(x^2)) 2x \\ &= -2x \sin(x^2) e^{\cos(x^2)} \end{aligned}$$

Exponentials

- Note that $a^{b+c} = a^b a^c$ (that is we multiply a together $b + c$ times)

- Now $e^\epsilon \approx (1 + \epsilon)$



- But $e^{x+\epsilon} = e^x e^\epsilon = e^x (1 + \epsilon + O(\epsilon^2)) = e^x + \epsilon e^x + O(\epsilon^2)$

$$\frac{de^x}{dx} = \lim_{\epsilon \rightarrow 0} \frac{e^{x+\epsilon} - e^x}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon e^x + O(\epsilon^2)}{\epsilon} = e^x$$

Natural Logarithms

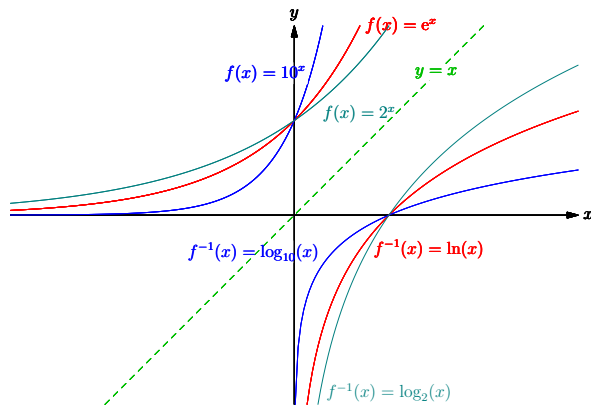
- The natural logarithm is defined as the inverse of e^x

$$\ln(e^x) = x \quad e^{\ln(y)} = y$$

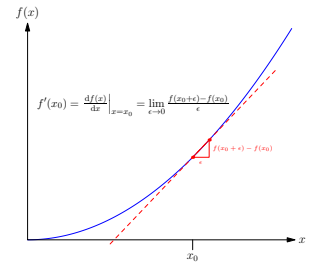
- Recall that if $g(y) = f^{-1}(y)$ then $g'(y) = 1/f'(g(y))$

- Consider $g(y) = \ln(y)$ and $f(x) = e^x$ (with $f'(x) = e^x$)

$$\frac{d\ln(y)}{dy} = \frac{1}{e^{\ln(y)}} = \frac{1}{y}$$



1. Why Calculus?
2. Differentiation
3. **Vector Calculus**
4. Matrix Calculus



Derivatives in High Dimensions

- When working with functions $f: \mathbb{R}^n \rightarrow \mathbb{R}$ in many dimensions then there will typically be different derivative in different directions
- To compute the derivative in a direction $\mathbf{u} \in \mathbb{R}^n$ (where $\|\mathbf{u}\| = 1$) at a point $\mathbf{x} \in \mathbb{R}^n$ we use

$$\partial_{\mathbf{u}} F(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \mathbf{u}) - f(\mathbf{x})}{\epsilon}$$

- If $\mathbf{u} = \delta_i = (0, \dots, 0, 1, 0, \dots, 0)^T$ (i.e. $u_i = 1$) then

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \delta_i) - f(\mathbf{x})}{\epsilon}$$

Computing Gradients 1

- We can compute the gradient by writing out $f(\mathbf{x})$ componentwise and performing the partial derivative with respect to x_i

$$\begin{aligned} \nabla \mathbf{w}^T \mathbf{M} \mathbf{w} &= \begin{pmatrix} \frac{\partial}{\partial w_1} \\ \frac{\partial}{\partial w_2} \\ \frac{\partial}{\partial w_3} \\ \vdots \end{pmatrix} \sum_{i,j} w_i M_{ij} w_j = \begin{pmatrix} \sum_j M_{1j} w_j + \sum_i w_i M_{i1} \\ \sum_j M_{2j} w_j + \sum_i w_i M_{i2} \\ \sum_j M_{3j} w_j + \sum_i w_i M_{i3} \\ \vdots \end{pmatrix} \\ &= \mathbf{M} \mathbf{w} + \mathbf{M}^T \mathbf{w} = (\mathbf{M} + \mathbf{M}^T) \mathbf{w} \end{aligned}$$

- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

Function Composition

- The gradient tells us about the change in a function value when we perturb an input $\mathbf{x}$ by a small amount in the directions $\mathbf{u}$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^T \nabla f(\mathbf{x}) + O(\epsilon^2)$$

- What happens when we compose functions: $h(\mathbf{x}) = f(g(\mathbf{x}))$? Note that,

$$\mathbf{x} \in \mathbb{R}^n \quad \mathbf{g}: \mathbb{R}^n \rightarrow \mathbb{R}^m \quad \mathbf{f}: \mathbb{R}^m \rightarrow \mathbb{R} \quad \mathbf{h}: \mathbb{R}^n \rightarrow \mathbb{R}$$

- The vector valued function $\mathbf{g}(\mathbf{x})$ can be seen as m independent

$$\text{function } \mathbf{g}(\mathbf{x}) = \begin{pmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_m(\mathbf{x}) \end{pmatrix}$$

Taylor

- If we expand $f(\mathbf{x} + \epsilon \mathbf{u})$ to first order in ϵ

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^T \mathbf{g}(\mathbf{x}) + O(\epsilon^2)$$

$$\text{then } g_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}$$

- Recall we defined the vector of first order derivatives of $f(\mathbf{x})$ to be the gradient

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

- Thus

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^T \nabla f(\mathbf{x}) + O(\epsilon^2)$$

This is the start of the high-dimensional Taylor expansion

Computing Gradients 2

- A slicker way is just to expand $f(\mathbf{x} + \epsilon \mathbf{u})$

- Consider $f(\mathbf{x}) = \mathbf{x}^T \mathbf{M} \mathbf{x} + \mathbf{a}^T \mathbf{x}$

$$\begin{aligned} f(\mathbf{x} + \epsilon \mathbf{u}) &= (\mathbf{x} + \epsilon \mathbf{u})^T \mathbf{M} (\mathbf{x} + \epsilon \mathbf{u}) + \mathbf{a}^T (\mathbf{x} + \epsilon \mathbf{u}) \\ &= f(\mathbf{x}) + \epsilon (\mathbf{u}^T \mathbf{M} \mathbf{x} + \mathbf{x}^T \mathbf{M} \mathbf{u} + \mathbf{a}^T \mathbf{u}) + O(\epsilon^2) \\ &= f(\mathbf{x}) + \epsilon \mathbf{u}^T (\mathbf{M} \mathbf{x} + \mathbf{M}^T \mathbf{x} + \mathbf{a}) + O(\epsilon^2) \end{aligned}$$

$$\text{using } \mathbf{x}^T \mathbf{M} \mathbf{u} = \mathbf{u}^T \mathbf{M}^T \mathbf{x} \text{ and } \mathbf{a}^T \mathbf{u} = \mathbf{u}^T \mathbf{a}$$

- But $f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^T \nabla f(\mathbf{x}) + O(\epsilon^2)$ so

$$\nabla f(\mathbf{x}) = \mathbf{M} \mathbf{x} + \mathbf{M}^T \mathbf{x} + \mathbf{a}$$

Jacobians

- Perturbing $\mathbf{x}$ by $\epsilon \mathbf{u}$

$$g_i(\mathbf{x} + \epsilon \mathbf{u}) = g_i(\mathbf{x}) + \epsilon \mathbf{u}^T \nabla g_i(\mathbf{x}) + O(\epsilon^2)$$

so that

$$\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u}) = \mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u} + O(\epsilon^2)$$

where $\mathbf{J}$ is an $m \times n$ matrix with components $J_{ij} = \frac{\partial g_i(\mathbf{x})}{\partial x_j}$

- Note that $\mathbf{J} \mathbf{u}$ is a column vector of size m
- Ignoring terms of order ϵ^2

$$\begin{aligned} h(\mathbf{x} + \epsilon \mathbf{u}) &= f(\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u})) = f(\mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u}) \\ &= f(\mathbf{g}(\mathbf{x})) + \epsilon \mathbf{u}^T \mathbf{J}^T \nabla f(\mathbf{g}(\mathbf{x})) \end{aligned}$$

- Using $h(x) = f(g(x))$

$$h(x + \epsilon u) = h(x) + \epsilon u^T J^T \nabla_g f(g(x))$$

- We get that

$$\nabla_x h(x) = \nabla_x f(g(x)) = J^T \nabla_g f(g(x))$$

- Componentwise

$$\frac{\partial h(x)}{\partial x_i} = \frac{\partial f(g(x))}{\partial x_i} = \sum_{j=1}^m \frac{\partial g_j(x)}{\partial x_i} \frac{\partial f(g(x))}{\partial g_j}$$

Differentiating Matrices

- Often we have loss functions with respect to a matrix W , e.g.

$$L(W) = (a^T W b - c)^2$$

- We might want to find the minimum with respect to W
- This occurs at a point W^* where $L(W)$ does not increase as we change W in any way
- That is, we seek a W^* such that, for any matrices U

$$L(W^* + \epsilon U) - L(W^*) = O(\epsilon^2)$$

Example

- Suppose

$$L(W) = (a^T W b - c)^2$$

then

$$\begin{aligned} L(W + \epsilon U) &= (a^T (W + \epsilon U) b - c)^2 = (a^T W b + \epsilon a^T U b - c)^2 \\ &= L(W) + 2\epsilon (a^T W b - c) (a^T U b) + O(\epsilon^2) \end{aligned}$$

- Now

$$a^T U b = \sum_{ij} a_i U_{ij} b_j = \sum_{ij} U_{ji} a_j b_i = \text{tr } U^T a b^T$$

$$\text{Thus } \frac{\partial L(W)}{\partial W} = 2(a^T W b - c) a b^T$$

Quick Matrix Differentiation

- Let

$$\partial_U f(X) = \lim_{\epsilon \rightarrow 0} \frac{f(X + \epsilon U) - f(X)}{\epsilon} = \text{tr } U^T \frac{\partial f(X)}{\partial X}$$

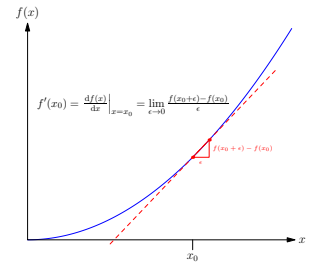
- E.g.

$$\begin{aligned} \partial_U \text{tr } A X B &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{tr } A (X + \epsilon U) B - \text{tr } A X B) \\ &= \text{tr } A U B = \text{tr } B^T U^T A^T = \text{tr } U^T A^T B^T \end{aligned}$$

thus

$$\frac{\partial \text{tr } A X B}{\partial X} = A^T B^T$$

- Why Calculus?
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Generalised Gradient

- We can generalise the idea of gradient to matrices

$$\frac{\partial L(W)}{\partial W} = \begin{pmatrix} \frac{\partial L(W)}{\partial W_{11}} & \frac{\partial L(W)}{\partial W_{12}} & \dots & \frac{\partial L(W)}{\partial W_{1m}} \\ \frac{\partial L(W)}{\partial W_{21}} & \frac{\partial L(W)}{\partial W_{22}} & \dots & \frac{\partial L(W)}{\partial W_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L(W)}{\partial W_{n1}} & \frac{\partial L(W)}{\partial W_{n2}} & \dots & \frac{\partial L(W)}{\partial W_{nm}} \end{pmatrix}$$

- From an identical argument we used for vectors

$$L(W + \epsilon U) = L(W) + \epsilon \text{tr } U^T \frac{\partial L(W)}{\partial W} + O(\epsilon^2)$$

where

$$\text{tr } U^T G = \sum_i [U^T G]_{ii} = \sum_{ij} U_{ji} G_{ji} = \sum_{ij} U_{ij} G_{ij} = \langle U, G \rangle$$

Traces

- The trace of a matrix is the sum of its diagonal elements

$$\text{tr } A = \text{tr } A^T = \sum_i A_{ii}$$

- Clearly $\text{tr } cA = c \text{tr } A$

- Also $\text{tr } (A + B) = \text{tr } A + \text{tr } B$

- We note that

$$\text{tr } AB = \sum_{i,j} A_{ij} B_{ji} = \sum_{i,j} B_{ij} A_{ji} = \text{tr } BA$$

- It follows that

$$\text{tr } ABCD = \text{tr } DABC = \text{tr } CDAB = \text{tr } BCDA$$

Summary

- With care you can differentiate most expressions
- The chain and product rule are incredibly powerful tools
- We can generalise differentiation to vectors and matrices
- There are a number of surprisingly useful results see [The Matrix Cookbook](#)
- When we look at [integration](#) it gets harder